When Heat Meets Health-System Capacity: A Reproducible Ecological Study of Outpatient Utilization Across China

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Abstract

Humid heat may affect healthcare demand while healthcare capacity may shape how health systems accommodate that demand. This study uses publicly auditable data for 31 mainland Chinese provincial units in 2014, 2016, and 2018 (93 province-year observations) to examine whether the association between warm-season Humidex and outpatient visits per resident differs by lagged healthcare capacity. Population-weighted ERA5 grid data, WorldPop population rasters, and official health statistics are combined in a reproducible workflow. The Humidex–capacity interaction is 0.0021 (normal-approximation clustered 95% CI: $[-0.0167, 0.0208]$; $p = 0.8294$). Wild-bootstrap ($p = 0.8129$) and permutation ($p = 0.6349$) inference provide no precise evidence of effect modification. The results are province-year ecological associations, not individual-level short-term effects or causal effects of resource allocation. **Keywords:** humid heat; Humidex; outpatient utilization; healthcare capacity; ecological study; China

1 Introduction

Climate change is increasingly a health-system question. Heat can trigger physiological strain, but the burden ultimately becomes visible through a chain that includes exposure, symptoms, care seeking, clinical triage, and the ability of local services to respond. Humidity matters along this chain because it impedes evaporative cooling and can intensify thermal discomfort (Masterton and Richardson, 1979; Raymond et al., 2020). Global evidence has established that hazardous heat is already consequential for population health (Mora et al., 2017; Gasparrini et al., 2015; Vicedo-Cabrera et al., 2021), while climate–health assessments increasingly foreground adaptation capacity alongside exposure (Romanello et al., 2023).

Yet the empirical evidence base remains uneven. Much of the strongest heat–health literature uses daily deaths, emergency presentations, or diagnosis-specific records; for example, multi-site work in China links high temperature to emergency-department use (Wang et al., 2020). These designs are indispensable for estimating acute individual-level risks, but they do not directly reveal whether the service environment alters the observed relationship between climate stress and recorded utilization across regions.

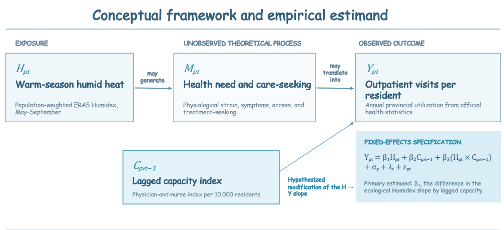

Figure 1: Conceptual framework.

We therefore ask a deliberately constrained systems question: at the province-year level, is the association between warm-season humid heat and outpatient visits systematically different in places with greater pre-existing healthcare capacity? The original project considered a person-level CFPS design, but the public files lack both exact visit dates and a comparable outpatient outcome. Rather than manufacture those missing links, we build a transparent ecological panel from open data. The contribution is a reproducible benchmark, a clearly scoped test of capacity modification, and an explicit accounting of what this design cannot identify.

Figure 1 summarizes the conceptual pathways.

2 Literature Review

Heat-health science has moved beyond temperature alone. Physiological theory and observational evidence both indicate that the joint temperature–moisture environment is central to heat stress (Masterton and Richardson, 1979; Sherwood and Huber, 2010; Raymond et al., 2020). Reanalysis products such as ERA5 make spatially consistent exposure measurement possible (Hersbach et al., 2020; Copernicus Climate Change Service, 2023), while population rasters can weight exposure toward where people live (WorldPop, 2023). These advances make it feasible to construct comparable large-area exposures, but they do not turn aggregated data into person-level outcomes.

Healthcare resources may operate as effect modifiers, but their direction is not theoretically fixed. Greater physician and nurse availability may accommodate demand that would otherwise be delayed or unmet; preventive and continuous care may instead avert some presentations. Recorded visits are consequently a joint signal of need and access, rather than a direct measure of disease burden. This ambiguity makes a signed interaction intrinsically insufficient as evidence of protection. We therefore foreground interval estimates, design diagnostics, and sensitivity analyses. With only 31 clusters, conventional clustered inference can also be unreliable, motivating complementary wild-bootstrap and randomization-based procedures (Cameron and Miller, 2015; MacKinnon and Webb, 2018).

3 Estimation Method

The main specification is

$$Y_{pt} = \beta_1 H_{pt} + \beta_2 C_{p,t-1} + \beta_3 H_{pt} C_{p,t-1} + \alpha_p + \lambda_t + \varepsilon_{pt}, \quad (1)$$

where Y_{pt} is outpatient visits per resident, H_{pt} is population-weighted warm-season Humidex, and $C_{p,t-1}$ is the preceding-year equal-weight healthcare-capacity index. Province and year fixed effects are denoted by α_p and λ_t. Standard errors are clustered by province. The interaction β_3 describes a difference in ecological slopes; it is not a causal parameter for healthcare-resource policy.

Because there are 31 clusters, the analysis reports normal-approximation clustered inference alongside Rademacher wild-bootstrap and province-label permutation inference. The latter preserves each province's within-series capacity structure while reassigning province series across units.

4 Data and Sample

The balanced sample contains 31 mainland provincial units in 2014, 2016, and 2018, yielding 93 observations. The outcome is annual outpatient visits per resident, constructed from official provincial outpatient tables. Physician and nurse inputs for the lagged capacity index come from official National Bureau of Statistics and National Health Commission materials (National Bureau of Statistics of China, 2015, 2017, 2019; National Bureau of Statistics of China and National Health Commission, 2014, 2016, 2018).

Daily 2-metre temperature and dew point from the Copernicus ERA5 0.25-degree product are converted to Humidex and aggregated over May–September with WorldPop population weights. This exposure is a province-wide population-weighted grid measure, not a provincial-centroid proxy. The capacity index averages sample-standardized physicians and nurses per 10,000 residents measured in the previous year.

Table 1 reports the annual descriptive statistics. Figure 2 displays geographic heterogeneity in the principal indicators.

Table 1: Descriptive statistics

Year	Outpatient visits per resident	Warm-season Humidex	Healthcare-capacity index
2014	5.4032	28.0258	-0.5088
2016	5.5916	28.9377	-0.0520
2018	5.7877	29.4420	0.5608

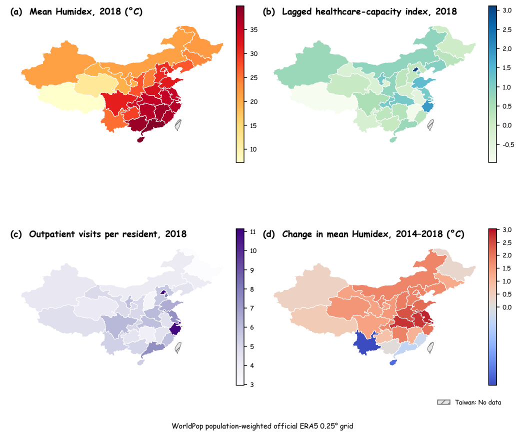

Figure 2: Provincial indicators in 2018. Humidex is a WorldPop population-weighted official ERA5 0.25-degree grid measure.

5 Empirical Results

5.1 A national, population-weighted exposure–capacity panel

The panel reveals substantial geographic variation in both humid heat and outpatient use (Table 1; Figure 2). This variation is analytically useful but should not be overinterpreted: a province's annual outpatient count aggregates heterogeneous populations, institutions, and clinical conditions. Our estimand is therefore a within-province, between-year ecological association, after removing time-invariant provincial differences and common year shocks.

5.2 No precise evidence that capacity modifies the association

Table 2 reports the main model. The estimated Humidex × capacity coefficient is 0.0021 with a clustered standard error of 0.0096. Under normal-approximation clustered inference, the 95% interval is $[-0.0167, 0.0208]$ and the p value is 0.8294. Wild-bootstrap and permutation p values are 0.8129 and 0.6349. A Stata implementation using 30 finite cluster degrees of freedom gives $p = .831$ and a CI of $[-.0175, .0216]$; this small difference reflects the inference approximation, not a different dataset or model.

The intervals span negative, positive, and near-zero associations. Thus, the central result is not evidence of no heat-related health burden, nor evidence that capacity is irrelevant. It is a more limited finding: this three-period ecological panel cannot distinguish plausible positive, negative, or negligible capacity modification slopes. Figure 3 makes that uncertainty visible; fitted relationships are close enough that their apparent ordering should not be read as a stable service-system mechanism.

Table 2: Main model results

Statistic	Estimate
Humidex × capacity	0.0021
Clustered standard error	0.0096
Normal-approximation clustered 95% CI	[-0.0167, 0.0208]
Observations	93
Provinces	31
R^2	0.9929
Normal-approximation clustered p value	0.8294
Wild-bootstrap p value	0.8129
Permutation p value	0.6349

Note: wild-bootstrap and permutation values provide alternative inference for the same interaction.

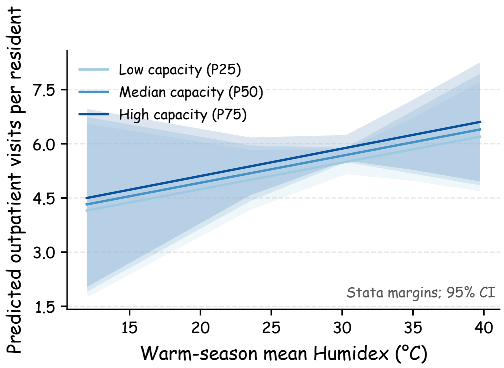

Figure 3: Fixed-effects interaction predictions.

Figure 3 visualizes fitted relationships at selected capacity values.

5.3 Supplementary Analysis: the null interaction is not driven by one operational definition

Sensitivity analyses use warm-season days with Humidex at least 35°C instead of mean Humidex; separate physician, nurse, and principal-component capacity measures; exclusion of the four municipalities; and leave-one-province-out estimation. These alternatives do not yield a directionally consistent and precise interaction pattern. The leave-one-out estimates are not driven by a single province, although that diagnostic cannot resolve measurement error or limited statistical power.

Figure 4 reports leave-one-province-out estimates, and Figure 5 shows observed common support.

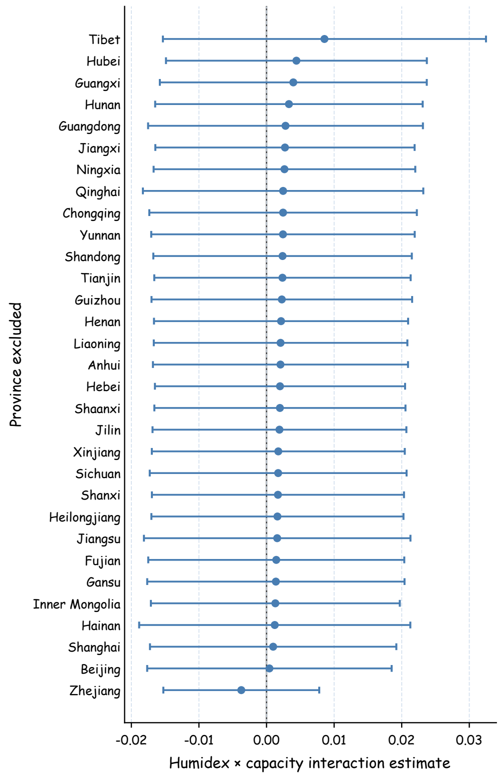

Figure 4: Leave-one-province-out interaction estimates.

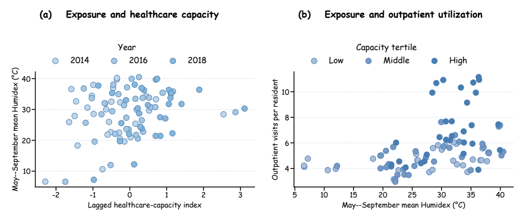

Figure 5: Common support for warm-season exposure and healthcare capacity.

6 Discussion

The results clarify a distinction that is easy to blur in climate–health research. The wider literature provides strong evidence that heat can damage health, and health systems are central to adaptation (Gasparrini et al., 2015; Vicedo-Cabrera et al., 2021; Romanello et al., 2023). Our analysis does not revisit that causal question. Instead, it tests whether a coarse, publicly reproducible provincial panel can resolve a specific service-capacity interaction. It cannot do so precisely.

That restraint is informative. Annual outpatient utilization is generated jointly by clinical need, affordability, referral practices, facility availability, and reporting systems. A capacity index based on physicians and nurses captures only one part of readiness; it cannot measure cooling infrastructure, surge protocols, quality, primary-care continuity, or household access. Province averages likewise cannot identify individual exposure, indoor conditions, diagnosis, or actual healthcare-seeking. Fixed effects remove time-invariant provincial differences and common year shocks, but not time-varying confounding.

The next empirical step is therefore not a stronger causal claim from the same aggregation. It is data linkage: daily, diagnosis-resolved utilization records; precise exposure assignment; measures of facility operations and access; and a pre-specified identification strategy. Such data could connect the acute-risk evidence from time-series studies to the institutional adaptation question that motivates this paper.

7 Conclusions

Using fully public climate, population, and health-statistics inputs, this paper constructs a reproducible province-year ecological panel for China. The main interaction estimate is small relative to its uncertainty, and conventional, wild-bootstrap, and permutation inference all fail to provide precise evidence of effect modification. The result places an empirical boundary around the present data rather than around the reality of climate-related health risk: the available panel supports transparent ecological description, but not individual-level causal claims. Future work should connect daily clinical records, precise place and time information, and health-system operations in a pre-specified quasi-experimental design.

Reproducibility Statement

Tables are generated from validated model-output CSV files with the scripts in this package. Figures are package-local analysis outputs. The manuscript can be rebuilt without network access after the local TeX environment is installed.

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